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SCHOOL OF COMPUTING AND ENGINEERING
Department of Computer Science

CMI3507-2425 (Data Mining) Assignment 2 – Evaluative Report

UNSUPERVISED ANALYSIS OF GLOBAL INFANT MORTALITY TRENDS

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Abstract

This report presents an analysis of global infant mortality data collected between 1960 and 2021. Initially, data preparation and exploratory data analysis are used to identify trends, cross-country patterns, and disparities in infant mortality outcomes. Unsupervised learning techniques are then applied, using K-means clustering, to group countries based on similarities in long-term infant mortality levels and variability. Clustering selection is assessed using the elbow method and silhouette scores to determine an appropriate number of clusters, and the results show distinct groups of countries with different mortality profiles, reflecting differences in development outcomes.

1 Introduction

Infant mortality rate (number of deaths of infants under one year, per 1,000 live births) is primarily used as an indicator of population health, socio-economic development, and effectiveness of healthcare systems. Analysis of infant mortality over long time periods provides valuable insight into global health progress and inequalities.

2 General Questions

2.1 Research Questions Formulation

Based on the structure and scope of the dataset, two research questions are formulated below:

1. How have infant mortality rates evolved across different countries, and what key temporal and cross-country patterns can be identified between 1960 and 2021?
2. Can countries be grouped into meaningful clusters based on similarities in infant mortality levels and trends, and what distinguishes these clusters?

2.2 Data Preparation –

The dataset structure was prepared to allow proper analysis. The primary dataset (T1) contains infant mortality rates for many countries, while an auxiliary dataset (T2) provides contextual information — regional classification and income group for each country.

The T1 dataset was originally in wide format, with yearly infant mortality rates stored as separate columns. This dataset was reshaped into long format so that each row represents a single country-year observation, to allow for temporal analysis and aggregation. The T2 dataset was subsequently merged using country names as a common key to add categorical variables that support comparative analysis.

2.2.1 Preparation of Data

2.2.1.1 Data Reshaping to Long Format

```
# change to long format
df_long = df.melt(
    id_vars=['country', 'country_code'],
    value_vars=year_cols,
    var_name='year',
    value_name='infant_mortality_rate'
)
```

Figure 1: A Python code snippet showing reshaping the data to long format.

	country	country_code	year	infant_mortality_rate
0	United Arab Emirates	ARE	1960	135.5
1	Antigua and Barbuda	ATG	1960	62.9
2	Australia	AUS	1960	20.3
3	Austria	AUT	1960	37.3
4	Belgium	BEL	1960	29.4

Figure 2: First five rows of the infant mortality dataset after reshaping into long format.

2.2.1.2 Dataset Integration

```
df_long = df_long.merge(
    t2[["region", "country", "income_group"]],
    on="country",
    how="left")
```

Figure 3: A Python code snippet showing the datasets being integrated.

	country	country_code	year	infant_mortality_rate	region	income_group
0	United Arab Emirates	ARE	1960	135.5	NaN	NaN
1	Antigua and Barbuda	ATG	1960	62.9	NaN	NaN
2	Australia	AUS	1960	20.3	East Asia & Pacific	High income
3	Austria	AUT	1960	37.3	Europe & Central Asia	High income
4	Belgium	BEL	1960	29.4	Europe & Central Asia	High income

Figure 4: First five rows of the infant mortality dataset after integration with the auxiliary dataset.

Figure 4 indicates there are missing values; this will be dealt with appropriately in the following section.

2.3 Basic EDA

The infant mortality dataset was explored to understand its structure, identify patterns and trends, and detect potential anomalies before more advanced analysis.

2.3.1 Size of the Dataset and Number of Variables

2.3.1.1 Size of the Dataset

```
df_long.size
```

```
43524
```

Figure 5: A Python code snippet showing the size of the data.

Figure 5 indicates that the size of the data is 43524.

2.3.1.2 Number of Variables

```
len(df_long)
```

```
7254
```

```
len(df_long.columns.unique())
```

```
6
```

Figure 6: A Python code snippet showing the number of rows and columns in the data.

Figure 6 shows that there are 7254 rows and 6 features in the data.

2.3.2 Data Types of Variables

The variables are listed below with their respective data types:

- country: categorical (string)
- country_code: categorical (string)
- year: numerical (integer)
- infant_mortality_rate: numerical (continuous, floating-point)
- region: categorical (string)
- income_group: categorical (string)

These data types are appropriate for both descriptive statistical analysis and subsequent modelling tasks.

2.3.2 Missing Data and Cleaning

2.3.2.1 Identification of Missing Values

```
df_long.isna().sum()
```

	0
country	0
country_code	0
year	0
infant_mortality_rate	0
region	4340
income_group	4340

Figure 7: A Python code snippet showing the number of missing values in each column.

Figure 7 shows no missing values for the majority of the variables, indicating the primary data for analysis is complete. Missing values were identified for the region and income_group columns due to countries included in the infant mortality dataset not being present in the auxiliary metadata dataset from which these attributes were obtained. As these variables are not used for modelling, the missing values are retained and excluded only from analyses that require regional or income-based comparisons.

2.3.3 Descriptive Statistical Analysis

Descriptive statistics were computed for the infant mortality rate variable, including mean, median, standard deviation, minimum, and maximum values.

2.3.3.1 Descriptive Statistics by Country

```python
country_stats = (
    df_long
    .groupby('country')['infant_mortality_rate']
    .agg(['mean', 'median', 'std', 'min', 'max'])
)

country_stats.head()
```

country	mean	median	std	min	max
Algeria	70.946774	43.35	49.873459	19.2	147.7
Antigua and Barbuda	16.659677	11.95	13.351373	5.2	62.9
Australia	9.341935	7.35	5.735754	3.2	20.3
Austria	12.125806	7.80	10.028140	2.9	37.3
Bahrain	30.591935	19.45	31.882851	5.9	133.0

Figure 8: A Python code snippet and table showing descriptive statistics of infant mortality by country.

Figure 8 provides statistics for five countries in the dataset, and shows a wide range in average and variance.

2.3.3.1 Descriptive Statistics by Year

```python
year_stats = (
    df_long
    .groupby('year')['infant_mortality_rate']
    .agg(['mean', 'median', 'std', 'min', 'max'])
)

year_stats.head()
```

year	mean	median	std	min	max
1960	98.122222	101.4	53.774106	16.3	220.4
1961	95.184615	95.2	53.096632	16.0	217.3
1962	92.407692	89.0	52.430937	15.6	214.3
1963	89.764957	86.1	51.801017	15.0	211.1
1964	87.294017	83.7	51.161253	14.3	208.1

Figure 9: A Python code snippet and table showing descriptive statistics of infant mortality by year.

Figure 8 shows an initial trend towards lower infant mortality, which will be investigated further.

3 Data-specific Questions

3.1 Infant Mortality Data Preprocessing and EDA

3.1.1 Main Measured Variable

The most important variable in the dataset is the infant mortality rate. All subsequent analysis in this section focuses on understanding the behaviour and distribution of this variable across countries and time.

3.1.2 Temporal Patterns and Trends

An examination of infant mortality rates over time reveals a clear overall downward trend across most countries between 1960 and 2021. This decline is evident when visualising country-level time series. While the general direction of change is consistent, the rate of decline varies substantially between countries.

Earlier decades are characterised by relatively high infant mortality rates, particularly in low- and middle-income countries. Over time, many countries exhibit steady and sustained reductions, whereas others show slower improvements or periods of stagnation. These temporal patterns highlight both global progress and persistent inequalities in child health outcomes.

3.1.2.2 Global Trends in Infant Mortality

```
global_trend = (
    df_long
    .groupby('year', as_index=False)['infant_mortality_rate']
    .mean()
)

plt.figure(figsize=(12, 8))

sns.lineplot(
    data=global_trend,
    x='year',
    y='infant_mortality_rate',
    linewidth=2
)

plt.xlabel('Year')
plt.ylabel('Mean Infant Mortality Rate (per 1,000 live births)')
plt.title('Global Mean Infant Mortality Rate Over Time')
```

Figure 10: Python code used to compute and visualise the global mean infant mortality rate over time.

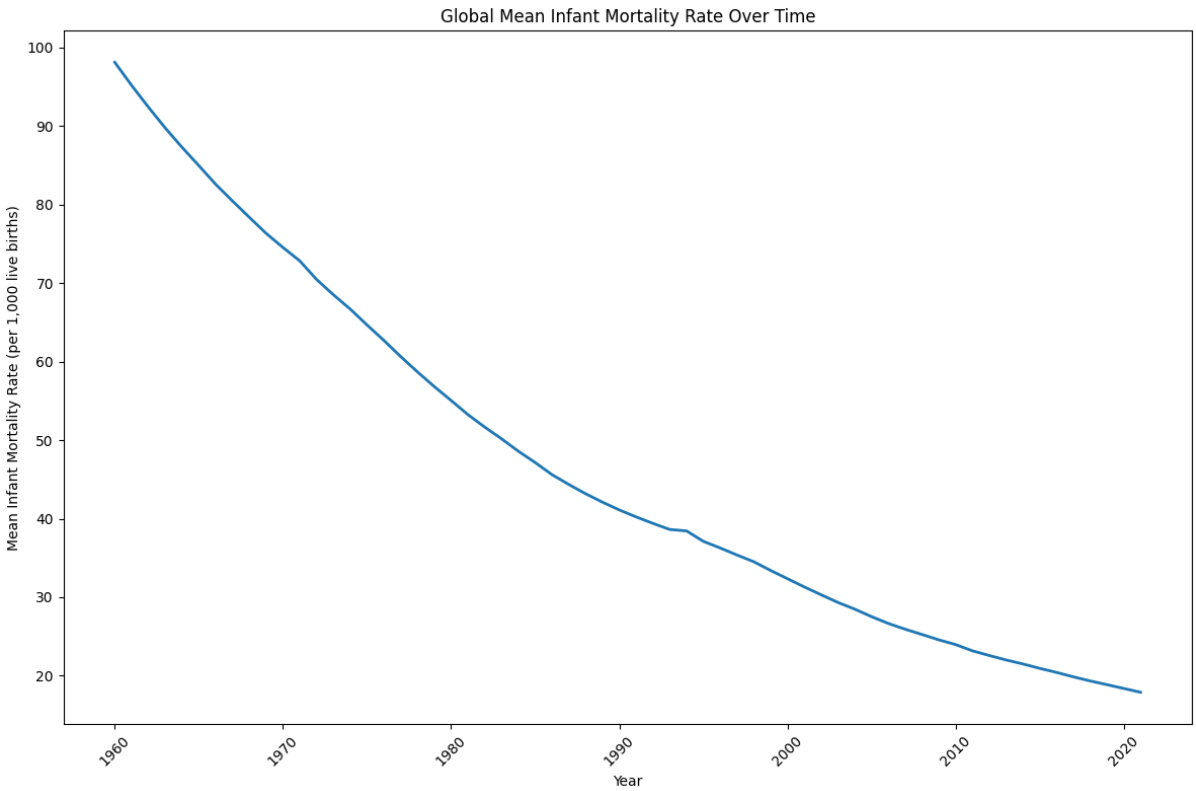

Figure 11: Global mean infant mortality rate over time, averaged across all countries in the dataset.

Figure 10 shows the Python code used to compute the global mean infant mortality rate, while Figure 11 illustrates the resulting trend toward lower rates.

3.1.3 Descriptive Statistics by Country and Region

The time series trend was analysed for random countries to highlight differences in outcomes. The results show countries with high rates tend to have better improvements, whereas other countries with lower rates have slower improvements, due to diminishing returns in healthcare improvements.

3.1.3.1 Temporal Trends in Infant Mortality for Selected Countries

```
np.random.seed(42)
random_countries = np.random.choice(
    df_long['country'].unique(),
    size=10,
    replace=False
)

df_subset = df_long[df_long['country'].isin(random_countries)]
```

Figure 12: Python code used to select 10 random countries, with a random seed for reproducibility.

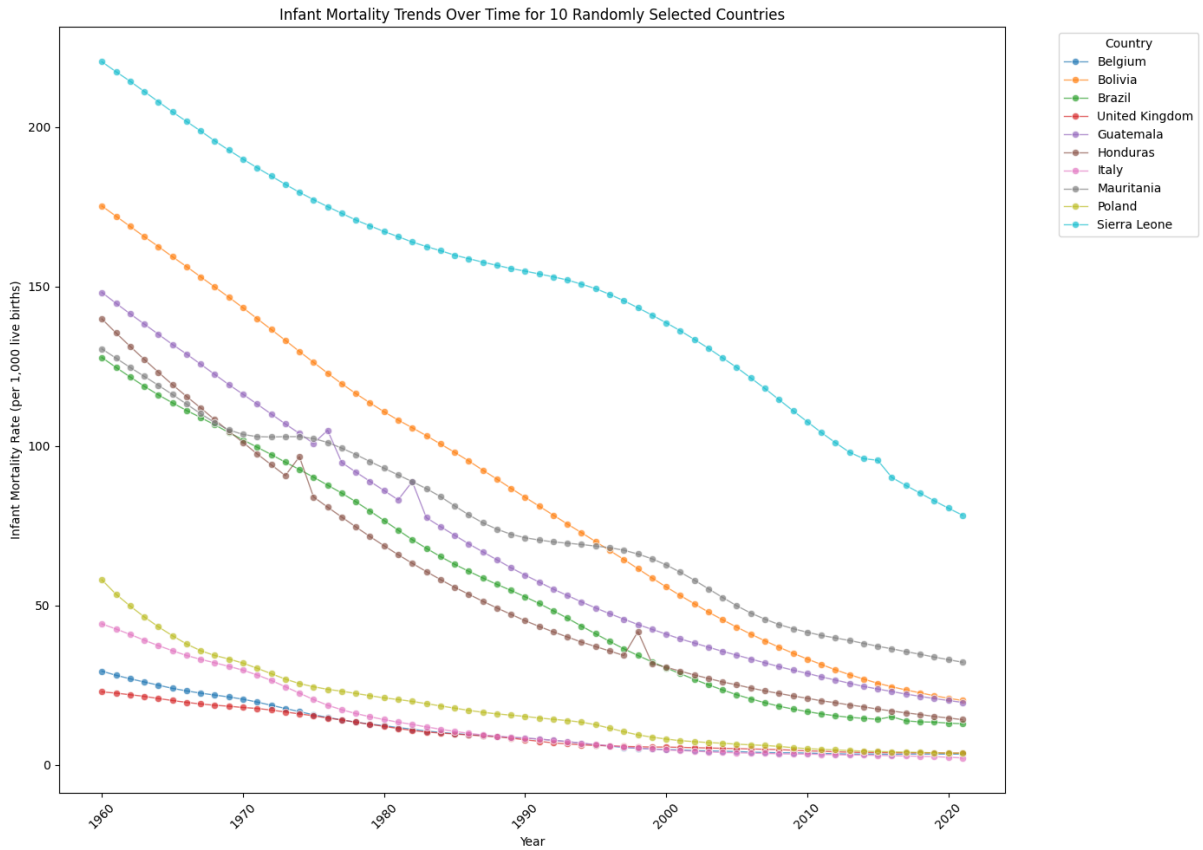

Figure 13: Infant mortality rate trends over time for 10 randomly selected countries, with connected markers indicating yearly observations.

3.1.3.2 Temporal Trends in Infant Mortality by Region

```
region_trends = (
    df_long.dropna(subset=['region'])
    .groupby(['year', 'region'])['infant_mortality_rate']
    .mean()
    .reset_index()
)
```

Figure 14: Python code used to differentiate regions.

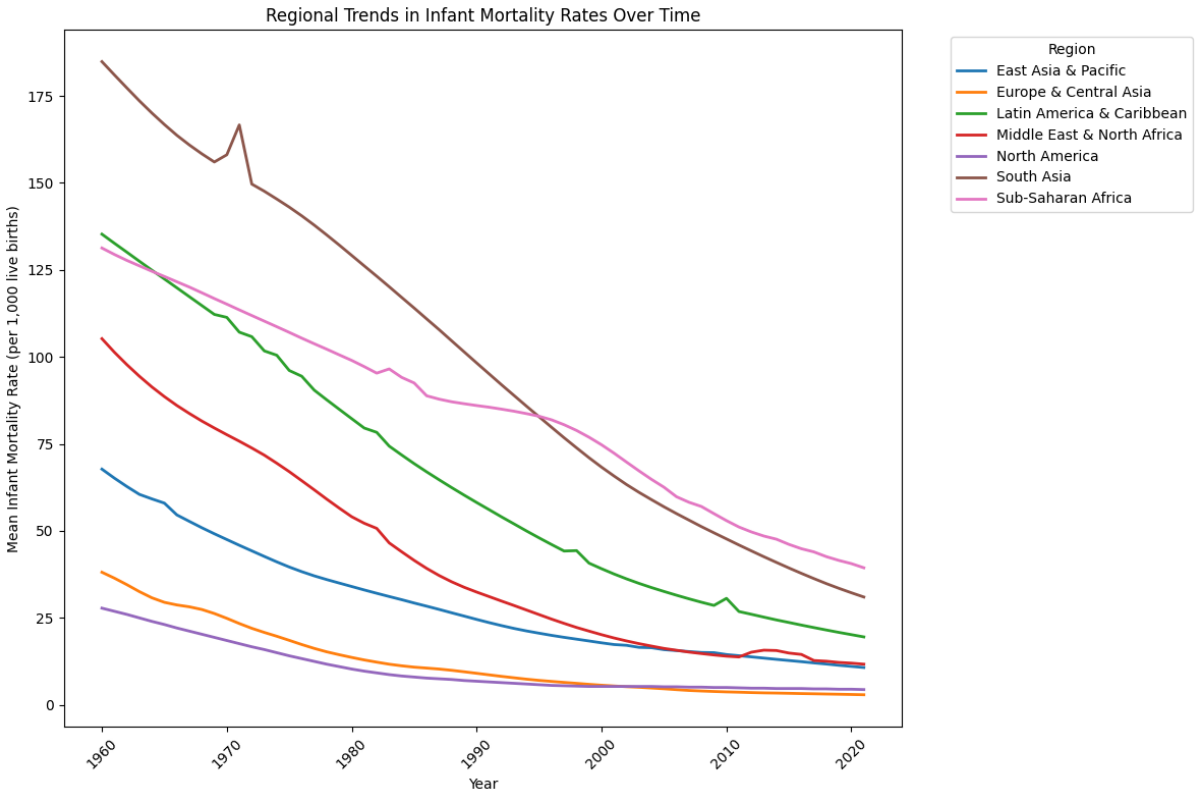

Figure 15: Infant mortality rate trends over time for each region.

Figure 15 highlights the differences in infant mortality rates between regions, showing global inequality between different areas of the world.

3.1.4 Outliers in Infant Mortality Rates

The analysis shows a greater presence of anomalies in the middle decades, particularly in the 1990s, possibly due to greater inequality. Considering Figure 15, there is stagnation in Sub-Saharan Africa, which could have contributed to these anomalies.

3.1.4.1 Examination of Outliers

```
plt.figure(figsize=(14, 8))

sns.boxplot(
    data=df_long,
    x='decade',
    y='infant_mortality_rate'
)
```

Figure 16: Python code used to construct boxplots of infant mortality rates by decade for outlier detection.

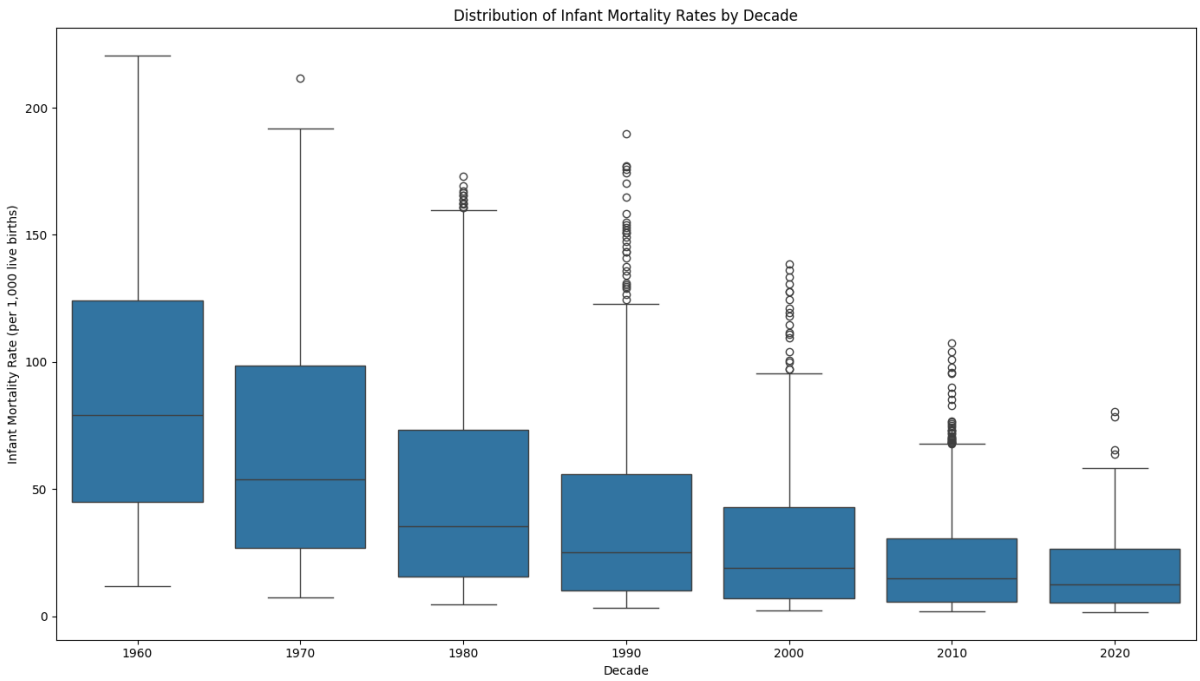

Figure 17: Boxplots showing the distribution of infant mortality rates by decade, highlighting outliers and changes in variability over time.

The boxplots reveal substantial outliers in infant mortality rates during the middle decades. Over time, both the median infant mortality rate and the overall spread of values decrease markedly, indicating not only global improvement but also a reduction in variability across countries. However, more extreme outlier suggest certain countries falling behind the global trend.

Q3.2: Model Building (Clustering) using the Infant Mortality Data –

Clustering was used as an unsupervised learning technique to identify countries with similar infant mortality characteristics. The K-means clustering algorithm was selected for efficiency. K-means partitions the dataset into k clusters by minimising within-cluster variance.

3.2.1 Data Preparation for Clustering

Long-term patterns were used as clustering features, such as the magnitude and variability of infant mortality over time, for comparison between countries.

3.2.1.1 Organisation of Features

```
country_features = (
    df_long
    .groupby('country')
    .agg(
        mean_rate=('infant_mortality_rate', 'mean'),
        median_rate=('infant_mortality_rate', 'median'),
        std_rate=('infant_mortality_rate', 'std'),
        min_rate=('infant_mortality_rate', 'min'),
        max_rate=('infant_mortality_rate', 'max')
    )
    .dropna()
)
```

Figure 18: Python code for organisation of features.

3.2.1.2 Standardisation of Features

All features were standardised using z-score normalisation.

```
from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(country_features)
```

Figure 19: Python code for standardisation.

3.2.2 Determining the Optimal Number of Clusters

3.2.2.1 Elbow Method

This method examines the relationship between the number of clusters and the total within-cluster sum of squares (inertia).

```python
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

for k in range(2, 9):
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X_scaled)
    inertia.append(km.inertia_)

plt.plot(range(2, 9), inertia, marker='o')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method for Optimal k')
plt.show()
```

Figure 20: Python code for the elbow method.

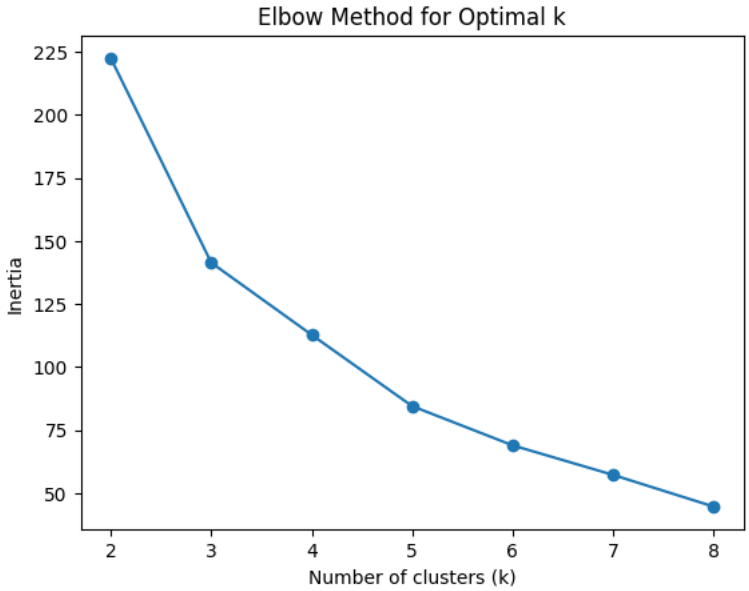

Figure 21: Graph for the Elbow Method.

Figure 21 shows diminishing returns beyond k=3.

3.2.2.2 Silhouette Score

This method assesses cluster separation and cohesion.

```
from sklearn.metrics import silhouette_score

silhouette_scores = []

for k in range(2, 9):
    km = KMeans(n_clusters=k, random_state=42)
    labels = km.fit_predict(X_scaled)
    silhouette_scores.append(silhouette_score(X_scaled, labels))

plt.plot(range(2, 9), silhouette_scores, marker='o')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Silhouette score')
plt.title('Silhouette Analysis for k Selection')
plt.show()
```

Figure 22: Python code for the silhouette score.

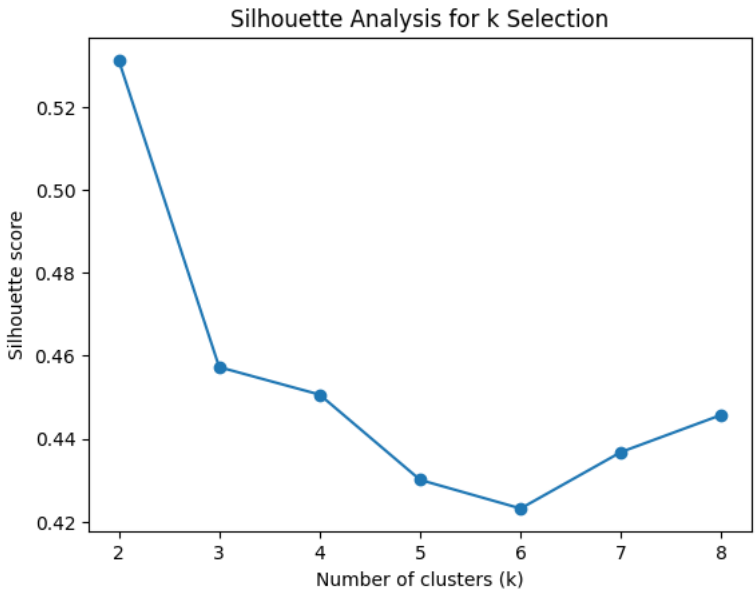

Figure 22: Graph for the silhouette score.

Figure 22 shows a relatively high score for three clusters, supporting the choice suggested by the elbow method.

3.2.3 Chosen Clustering Model

The K-means model was fitted with three clusters, as this was supported by the selection methods. The result is three distinct clusters of countries that represent different mortality profiles.

3.2.3.1 Model Creation

```
kmeans = KMeans(n_clusters=3, random_state=42)
country_features['cluster'] = kmeans.fit_predict(X_scaled)
```

Figure 23: Python code for model creation.

4 Literature Review

Unsupervised learning techniques are widely used in exploratory data analysis, often when labelled data is unavailable, to find patterns in complex datasets. Unlike supervised learning, unsupervised learning focuses on discovering structure, relationships, and similarities directly from the data. Common unsupervised learning tasks include clustering, dimensionality reduction, and association rule mining; these methods are especially valuable in healthcare and public health contexts.

4.1 Clustering as a Core Unsupervised Learning Task

Clustering groups data points with high similarity. Xu and Tian (2015) surveyed many clustering algorithms, with categories such as partition-based, hierarchical, density-based, grid-based, and model-based. Their review highlights that each clustering method has inherent strengths and limitations, and that algorithm selection should be guided by data characteristics such as size, dimensionality, and noise levels.

The K-means algorithm and other partition-based algorithms are often used for computational efficiency and ease of implementation. However, Xu and Tian (2015) note that K-means is sensitive to initial centroid selection, such as the mean or median infant mortality rate, and requires the number of clusters to be predefined. On the other hand, hierarchical clustering methods offer greater interpretability through dendrogram structures but can be computationally expensive for large datasets. Therefore, some pros and cons must be taken into account when analysing global health datasets such as infant mortality data, which span multiple decades and geographic regions.

4.2 Time-Series Clustering and Temporal Data

Infant mortality rates were represented as longitudinal data in this report, making time-series clustering a particularly suitable approach for exploratory analysis. Aghabozorgi, Shirkhorshidi, and Wah (2015) present a detailed review of time-series clustering techniques and emphasise their importance for discovering temporal patterns in datasets similar to the infant mortality dataset used. They also identified feature-based clustering as a viable approach for model creation, as this method is especially effective for large-scale datasets, by summarising each time-series using extracted features such as trends, averages, and variability. This approach improves scalability and

reduces sensitivity to noise and missing values, which are common issues in real-world health data. Aghabozorgi et al. (2015) also highlight the importance of appropriate distance measures and dimensionality reduction techniques when clustering time-series data, which shows the need for careful preprocessing before analysis.

4.3 Unsupervised Learning in Healthcare and Public Health

Unsupervised learning techniques have been widely adopted in healthcare analytics, as shown by Beard et al. (2015), who discuss the increasing role of big data analytics in healthcare and highlight clustering as a key method for identifying patient subgroups, disease patterns, and population health trends. They emphasise that unsupervised learning is particularly valuable in healthcare contexts as labelled outcomes are often limited or unavailable.

Other studies have demonstrated the application of clustering techniques to public health and mortality data. For example, Abdi and Williams (2018) show that clustering can be used to group countries based on health indicators, showing differences in outcomes across regions and income levels. Similarly, the World Health Organisation (2020) reports that it uses exploratory data analysis and clustering-based methods to examine global health inequalities and monitor long-term trends in mortality indicators. These studies support the use of unsupervised learning techniques for analysing infant mortality data at a global scale.

4.4 Professional, Ethical, and Social Considerations

Although the infant mortality data used in this study are aggregated and publicly available, clustering results must be interpreted with caution, as clusters do not imply causation, and there is a risk of oversimplifying complex socio-economic and political factors underlying health outcomes. Therefore, transparency in methodology and responsible interpretation are essential to avoid misleading conclusions, instead, actionable insights should be found to lower global inequality.

References

Abdi, H., & Williams, L. J. (2010). Principal component analysis. Wiley Interdisciplinary Reviews. Computational Statistics, 2(4), 433–459. https://doi.org/10.1002/wics.101

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Appendices

Appendix 2.1 - Python Code for Initial Setup

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("willianoliveiragibin/mortality-rate-infant-per-1000-live-births")

print("Path to dataset files:", path)

import pandas as pd
import os

os.listdir(path)

df = pd.read_csv(os.path.join(path, "T1.csv"), skiprows=1)

# rename columns
df = df.rename(columns={
    'Country Name': 'country',
    'Country Code': 'country_code',
    'Indicator Name': 'indicator',
    'Indicator Code': 'indicator_code'
})

year_cols = [c for c in df.columns if str(c).isdigit()]  # years like '1960'...'2021'
df = df[['country', 'country_code'] + year_cols]

# change to numeric
df[year_cols] = df[year_cols].apply(pd.to_numeric, errors='coerce')
```

Appendix 2.2 - Python Code for Question 2.2

```
# change to long format
df_long = df.melt(
    id_vars=['country', 'country_code'],
    value_vars=year_cols,
    var_name='year',
    value_name='infant_mortality_rate'
)
df_long['year'] = df_long['year'].astype(int)

t2 = pd.read_csv(os.path.join(path, "T2.csv"))

t2 = t2.rename(columns={
    'TableName': 'country',
    'IncomeGroup': 'income_group',
    'Region': 'region'
})

df_long = df_long.merge(
    t2[["region", "country", "income_group"]],
    on="country",
    how="left")
```

Appendix 2.3 - Python Code for Question 2.3

```
df_long.size
len(df_long)
len(df_long.columns.unique())
df_long.isna().sum()

country_stats = (
    df_long
    .groupby('country')['infant_mortality_rate']
    .agg(['mean', 'median', 'std', 'min', 'max'])
)

year_stats = (
    df_long
    .groupby('year')['infant_mortality_rate']
    .agg(['mean', 'median', 'std', 'min', 'max'])
)
```

Appendix 3.1 - Python Code for Question 3.1

```python
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np

global_trend = (
    df_long
    .groupby('year', as_index=False)['infant_mortality_rate']
    .mean()
)

plt.figure(figsize=(12, 8))

sns.lineplot(
    data=global_trend,
    x='year',
    y='infant_mortality_rate',
    linewidth=2
)

plt.xlabel('Year')
plt.ylabel('Mean Infant Mortality Rate (per 1,000 live births)')
plt.title('Global Mean Infant Mortality Rate Over Time')

# Decade ticks for cleanliness
decades = np.arange(
    global_trend['year'].min(),
    global_trend['year'].max() + 1,
    10
)
plt.xticks(decades, rotation=45)

plt.tight_layout()
plt.show()
```

```python
np.random.seed(42)
random_countries = np.random.choice(
    df_long['country'].unique(),
    size=10,
    replace=False
)

df_subset = df_long[df_long['country'].isin(random_countries)]

plt.figure(figsize=(14, 10))

sns.lineplot(
    data=df_subset,
    x='year',
    y='infant_mortality_rate',
    hue='country',
    marker='o',
    linewidth=1,
    alpha=0.7
)

plt.xlabel('Year')
plt.ylabel('Infant Mortality Rate (per 1,000 live births)')
plt.title('Infant Mortality Trends Over Time for 10 Randomly Selected Countries')

plt.xticks(decades, rotation=45)

plt.legend(
    title='Country',
    bbox_to_anchor=(1.05, 1),
    loc='upper left'
)

plt.tight_layout()
plt.show()
```

```
region_trends = (
    df_long.dropna(subset=['region'])
    .groupby(['year', 'region'])['infant_mortality_rate']
    .mean()
    .reset_index()
)

plt.figure(figsize=(12, 8))

sns.lineplot(
    data=region_trends,
    x='year',
    y='infant_mortality_rate',
    hue='region',
    linewidth=2
)

plt.xlabel('Year')
plt.ylabel('Mean Infant Mortality Rate (per 1,000 live births)')
plt.title('Regional Trends in Infant Mortality Rates Over Time')

plt.xticks(decades, rotation=45)

plt.legend(title='Region', bbox_to_anchor=(1.05, 1), loc='upper left')
plt.tight_layout()
plt.show()
```

```
df_long['decade'] = (df_long['year'] // 10) * 10

plt.figure(figsize=(14, 8))

sns.boxplot(
    data=df_long,
    x='decade',
    y='infant_mortality_rate'
)

plt.xlabel('Decade')
plt.ylabel('Infant Mortality Rate (per 1,000 live births)')
plt.title('Distribution of Infant Mortality Rates by Decade')

plt.tight_layout()
plt.show()
```

Appendix 3.2 - Python Code for Question 3.2

```python
country_features = (
    df_long
    .groupby('country')
    .agg(
        mean_rate=('infant_mortality_rate', 'mean'),
        median_rate=('infant_mortality_rate', 'median'),
        std_rate=('infant_mortality_rate', 'std'),
        min_rate=('infant_mortality_rate', 'min'),
        max_rate=('infant_mortality_rate', 'max')
    )
    .dropna()
)

from sklearn.preprocessing import StandardScaler

scaler = StandardScaler()
X_scaled = scaler.fit_transform(country_features)

from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

inertia = []

for k in range(2, 9):
    km = KMeans(n_clusters=k, random_state=42)
    km.fit(X_scaled)
    inertia.append(km.inertia_)

plt.plot(range(2, 9), inertia, marker='o')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Inertia')
plt.title('Elbow Method for Optimal k')
plt.show()

from sklearn.metrics import silhouette_score

silhouette_scores = []

for k in range(2, 9):
    km = KMeans(n_clusters=k, random_state=42)
    labels = km.fit_predict(X_scaled)
    silhouette_scores.append(silhouette_score(X_scaled, labels))

plt.plot(range(2, 9), silhouette_scores, marker='o')
plt.xlabel('Number of clusters (k)')
plt.ylabel('Silhouette score')
plt.title('Silhouette Analysis for k Selection')
plt.show()

kmeans = KMeans(n_clusters=3, random_state=42)
country_features['cluster'] = kmeans.fit_predict(X_scaled)
```